



RNAival

Martin-Luther-Universität Halle-Wittenberg,

Halle (Saale)

Manual

RNAival is a python-based program for the identification of siRNAs candidates and the assessment of the processing of dsRNAs.

Source: <https://github.com/MaxiSack/RNAival>

Contact: maximilian.sack@informatik.uni-halle.de

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Installation

Requirements

RNAival runs natively on Linux and Mac. It also runs in WSL2 on Windows 11 or newer.

Pre-Installation on Windows

Requirements:

- WSL2-capable Windows Version (Windows 11)
- Internet connection
- Administrator rights

Steps:

1. Ensure that Virtualisation is enabled for your computer:
 1. Open “Windows features” (Use Windows search bar)
 2. Find the features:
 1. “Virtual Machine Platform” (a subset of Hyper-V), sometimes called “Windows Hypervisor platform”
 2. “Windows Subsystem for Linux”
 3. Deactivate them and reboot the computer
 4. Activate them and reboot the computer
2. Install Windows Subsystem for Linux (WSL)
 1. Open a PowerShell (search for PowerShell in the Windows search field)
 2. Type `wsl --install` and hit Enter/Return
 - * This only works if WSL is not installed at all. If you get the WSL help text, i.e. an explanation of parameters and usage, a specific linux distribution needs to be specified: Use `wsl --list -online` to see which are available and install one of them (like “Ubuntu”, the default) with `wsl -install -d Ubuntu`
 3. Once prompted, set a username without spaces and a password
 4. Wait for the installation to finish
 5. Use `wsl --update` to ensure that the install contains all required features
 6. For more details, see <https://learn.microsoft.com/en-us/windows/wsl/install>

3. Using RNAival

1. Extract the program directory into a directory of your choice
2. Right-click in the directory containing the program (install.sh, run.sh)
3. Select “Open in terminal”

*This option may not be available on your windows installation.

In that case you will have to open a terminal and change the working directory to the program folder: `cd \path\to\directory`.

If the program folder is on a different drive (like D), use:

`cd /d D:\path\to\directory` instead.

4. Type `wsl` before all following commands or use `wsl` to turn the terminal into a linux terminal and then use all commands normally

Installation

1. Extract the program directory into a directory of your choice
2. Open a Terminal in the directory containing the program (install.sh)
3. Execute install.sh with `bash install.sh` (For windows: `wsl bash install.sh`)

This requires an internet connection and installs conda.

- By running this script, you implicitly accept miniforge’s license agreements (<https://github.com/conda-forge/miniforge/blob/main/LICENSE>).
- The program installs the conda dependencies into the home directory under “RNAival_Dependencies” (on windows it’s the wsl home directory).
- On Linux and Windows, the program also attempts to create a desktop entry in “/usr/share/applications” and requires the users password to do so (the WSL password on windows). This is required on Windows/WSL to add RNAival to the application list and make it available from windows. For Linux, not entering the password will instead create an entry locally (./local/share/applications).
- For WSL, the installer will also attempt to install the packages wslu and xdg-utils, which are only required for linking the Source and Contact web-pages and open the Manual (this file) from the program itself.

Source: <https://github.com/MaxiSack/RNAival>

Contact:

<https://www.informatik.uni-halle.de/arbeitsgruppen/bioinformatik/mitarbeiterinnen/sack/>

4. Wait for the installation to finish

Starting RNAival

- Find “RNAival” in the application menu of your operating system and launch the application
- OR (if you need more debugging output):
 1. Open a Terminal in the directory containing the program (run.sh)
 2. Execute install.sh with `bash run.sh` (For windows: `wsl bash run.sh`)

Uninstalling RNAival

1. Open a Terminal in the directory containing the program (uninstall.sh)
2. Execute uninstall.sh with `bash uninstall.sh` (For windows: `wsl bash uninstall.sh`)

Basic usage

- Create a new project (Menubar -> File -> New)
- Select sequencing libraries (fastq.gz)
- Select target sequences (embl,fasta,fna.gz)
- Select targets and evaluation types for libraries
- Run the pipeline

GUI overview

The following excerpts from the GUI are based on data from Knoblich *et al.* 2023¹. They investigated protecting plants against the cucumber mosaic virus (CMV) by using specially designed dsRNAs that interact with the plant's RNA interference pathway.

1. Knoblich, Marie, et al. "A new level of RNA-based plant protection: dsRNAs designed from functionally characterized siRNAs highly effective against Cucumber mosaic virus." *Nucleic Acids Research* 53.5 (2025): [gkaf136](#).

Input and Settings

Input files

1. Add the sequencing files and assign them appropriate names and comments
2. Add the mapping targets:
 1. Press "Add new target" and assign a name and optional comment
 2. Add the main sequence
 - Fasta for AGO-IP / siRNA identification
 - embl (annotated dsRNA) for dsRNA processing analysis
 3. Optionally add further background sequences (fasta, fna.gz)
3. Select the created targets for the corresponding libraries
 1. For each library with this target, select the target from the dropdown in the "Targets" column
4. Select the appropriate evaluation types for the libraries from the "Evaluation" column:
 1. siRNA candidate identification (siI) for AGO-IP libraries
 2. dsRNA processing evaluation (dsP) for Dicer/DCL-processed dsRNA libraries
5. If your data conforms to the recommended protocol, run the pipeline

The sequencing libraries from both dsRNA constructs are mapped to the dsCMV6-21o sequence, since dsCMV6-21 is just a subsequence of that. Mapping to the same sequence where possible allows for better comparisons and evaluation of the differences of the dsRNAs.

Select sequencing libraries				
Select directory		Select files		
Label	Comment	Pre-processing	Targets	Evaluation
Ago1_1	AGO1-enriched	sRP-Cut4	CMV_RNA2	siI
Ago1_2	AGO1-enriched	sRP-Cut4	CMV_RNA2	siI
Ago2_1	AGO2-enriched	sRP-Cut4	CMV_RNA2	siI
Ago2_2	AGO2-enriched	sRP-Cut4	CMV_RNA2	siI
Ago2_3	AGO2-enriched	sRP-Cut4	CMV_RNA2	siI
DCL_1	Control	sRP-Cut4	CMV_RNA2	siI
DCL_2	Control	sRP-Cut4	CMV_RNA2	siI
DCL_3	Control	sRP-Cut4	CMV_RNA2	siI
dsCMV6-21	Blunt	sRP-Cut4	dsCMV6-21o	dsP
dsCMV6-21o	2 nt overhang	sRP-Cut4	dsCMV6-21o	dsP

Select targets for mapping
Add new target
CMV_RNA2
Main target: CMV_RNA2 Background sequences:
Main length: 3051
dsCMV6-21o
Main target: siR6CMV21nt172 Background sequences:
Main length: 172
Annotations: 16

Save Changes
Run entire pipeline

Parameters for short-read processing (sRP)

If your data conforms to the recommended protocol, no further action is required here, otherwise:

1. Create a new parameterset and name it appropriately
2. Set parameters for:
 - Step 1: Adapter clipping
 - Step 2: Paired end merging

Important here are the sequencing adapters used and possible other modifications. The study from which this sample data is, had 4 extra nucleotides around RNA sequences, which need to be clipped after the adapters have been removed. See the "Cut 5'/3' nucleotides after adapter clipping" parameter was set to "4" each.

Step 3 and 4 generally do not need to be modified.

3. Select the newly created parameterset on the input page, column "Pre-processing", for all libraries that use it

Run the pipeline (Menubar -> Actions -> Run pipeline)

File	Edit	Actions	Help
Input selection	Program log	sR-processing	sil eval dsP eval Graphical output Textual output
Parameters for short-read processing			
Parameter Sets	Step 1: Cut adapters from reads		
SRP-cut4	Parameters for cutadapt:		
sRP	3' Adapter for forward reads: TGGAAATTCCTCGGGTGCCAGGAAGTCCAGTCAC		
New	3' Adapter for reverse reads: AGATCGGAAGAGCGTCGTGTAGGGAAAGAGTG		
	Minimum Adapter length: 5		
	Allow insertions or deletions: ☒ Yes ☐ No		
	Error rate: 0.2		
	Minimum Read length: 15		
	Discard reads without adapter: Yes		
	Cut 5' nucleotides after adapter clipping: 4		
	Cut 3' nucleotides after adapter clipping: 4		
	Extra parameters:		
	Reset to defaults		
	Step 2: Merge paired-end reads		
	Parameters for NGmerge:		
	Minimum read overlap: 10		
	Produce shortest read when merging: ☒ Yes ☐ No		
	Extra parameters:		
	Reset to defaults		
	Step 3: Map reads to the target RNA		
	Step 3.1: Create an index of the target RNA (using bowtie-build)		
	Extra parameters:		
	Reset to defaults		
	Step 3.2: Map the reads onto the index (using bowtie)		
	Align full reads: Yes		
	Extra parameters:		
	Reset to defaults		
	Step 4: Count reads		
	Minimum read length: 15		
	Maximum read length: 30		
	Reset to defaults		

Parameters for siRNA candidate identification (sil)

After the pipeline has finished processing the data, the program needs to know which libraries are siRNA enriched and which are used as control, as well as AGO identity:

1. Add a new enriched-control-pair by selecting the plus (+)
2. Give the new pair an appropriate label / name
3. Select the target + parameterset with which the libraries were processed
4. Select the enriched and control samples in the appropriate columns
5. Reload the graphics (Menubar -> Actions -> Reload graphics)

File Edit Actions Help

Input selection Program log sR-processing **sil eval** dsP eval Graphical output Textual output

siRNA candidate identification

Settings
siRNA length:
Strand: ☒ Forward ☒ Reverse ☐ Both

Select pairs of libraries

Label:	Enriched:	Control:
AGO1-IP	☒	☒
Target: CMV_RNA2 + sRP-Cut4	☒	☒
Region-start: 1	Ago1_1	DCL_1
Region-end: 3051	Ago1_2	DCL_2
		DCL_3
AGO2-IP	☒	☒
Target: CMV_RNA2 + sRP-Cut4	☒	☒
Region-start: 1	Ago2_1	DCL_1
Region-end: 3051	Ago2_2	DCL_2
	Ago2_3	DCL_3
☒		

Parameters for dsRNA processing evaluation (dsP)

Evaluation of dsRNA processing requires no user input, but the resulting graphs can be customised:

- Use the checkbox to the right of each graph type to select whether to generate them or not
- Select the range of lengths to display the length distribution for
- Select which lengths to show on read distribution and coverage plots
Multiple length can be selected at the same time by using a comma-separated list, i.e. "21,22,23,24"
- Select the lengths and their colour for the multicoverage-lineplot
The default colour scheme follows the okabe-ito palette
- The Y-axes of all graphs of the same plot can be synchronised to have the same minimum and maximum values
This can highlight absolute differences between libraries, but make libraries with fewer reads less visible
- The Y-axes for the length distribution graph can instead be scaled to percent of all reads, while the read distribution for a specific length as percentage of all reads of the selected length
- Select colours used to show annotated siRNAs in the various graphs

File Edit Actions Help			
Input selection Program log sR-processing sil eval dsP eval Graphical output Textual output			
dsRNA processing evaluation			
Length distribution		Coverage for individual lengths	
Minimum length 15		Lengths (comma separated) 21	
Maximum length 30		X-label Position	
X-label Read length (nt)		Y-label Coverage	
Y-label abundance Abundance			
Y-label percent Percent of all reads			
Abundance of annotated RNAs		Coverage for multiple lengths	
X-label esiRNAs		X-label Position	
Y-label abundance Abundance		Y-label Coverage	
Y-label percent Percent of 21nt reads			
Read distribution for specific length		Length Colour (hex)	
Lengths (comma separated) 21		20 #E69F00	
X-label 5' position		21 #56B4E9	
Y-label abundance Abundance		22 #009E73	
Y-label percent Percent of 21nt reads		23 #F0E442	
		24 #0072B2	
		25 #D55E00	
		+	
		Heatmap	
		Highlight esiRNAs	
		Highlight other phase	
		Minimum length 15	
		Maximum length 30	
		Colourscale midpoint percent 95	
		X-label 5' Position	
		Y-label Read length (nt)	
		Global settings	
		Synchronise Y-Axes between graphs	
		Y-Axes (where : Abundance Percent Both)	
		Annotation colours	
		Primary guide strand colour #44aaff	
		Primary passenger strand colour #3355cc	
		Secondary guide strand colour #88ff88	
		Secondary passenger strand colour #22aa22	

Graphical output

After the pipeline has finished, or whenever you reload the graphics, RNAival will generate graphics for the evaluation of the data.

siRNA identification

[Not yet implemented] RNAival will mark and write out the top 30 siRNA candidates.

Alternatively, you can [clear the selection and] select them manually.

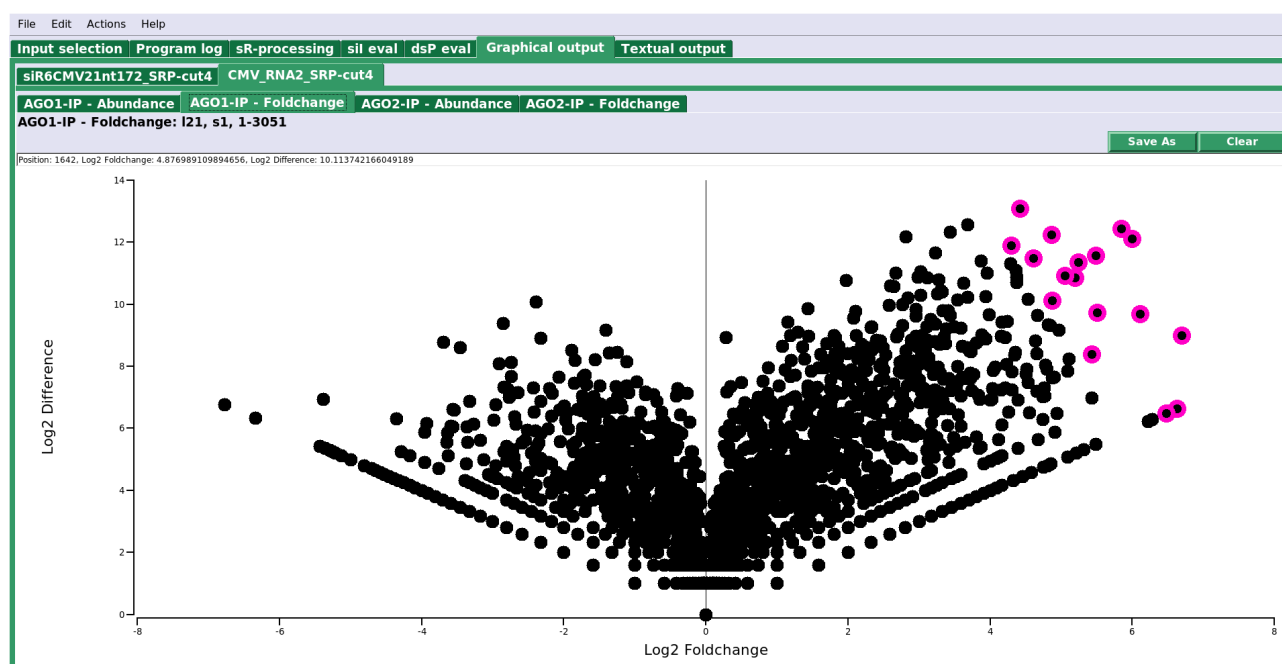
In the following example the top candidates have been manually selected.

Volcano-like plot

RNAival draws the (log2-) Difference in read counts against the (log2-) fold change of counts for each RNA of the specified length (default 21 nt), i.e. for each 5' position on the target RNA.

As a result, the best siRNA candidates are those RNAs that are most AGO conform, i.e. have both the highest fold change and difference in absolute abundance compared to the control sample (top-right, marked in pink).

This graph can be interacted with, allowing the user to select and of these RNAs and RNAival will subsequently output information on the selected RNAs in the Textual output tab (see also below).

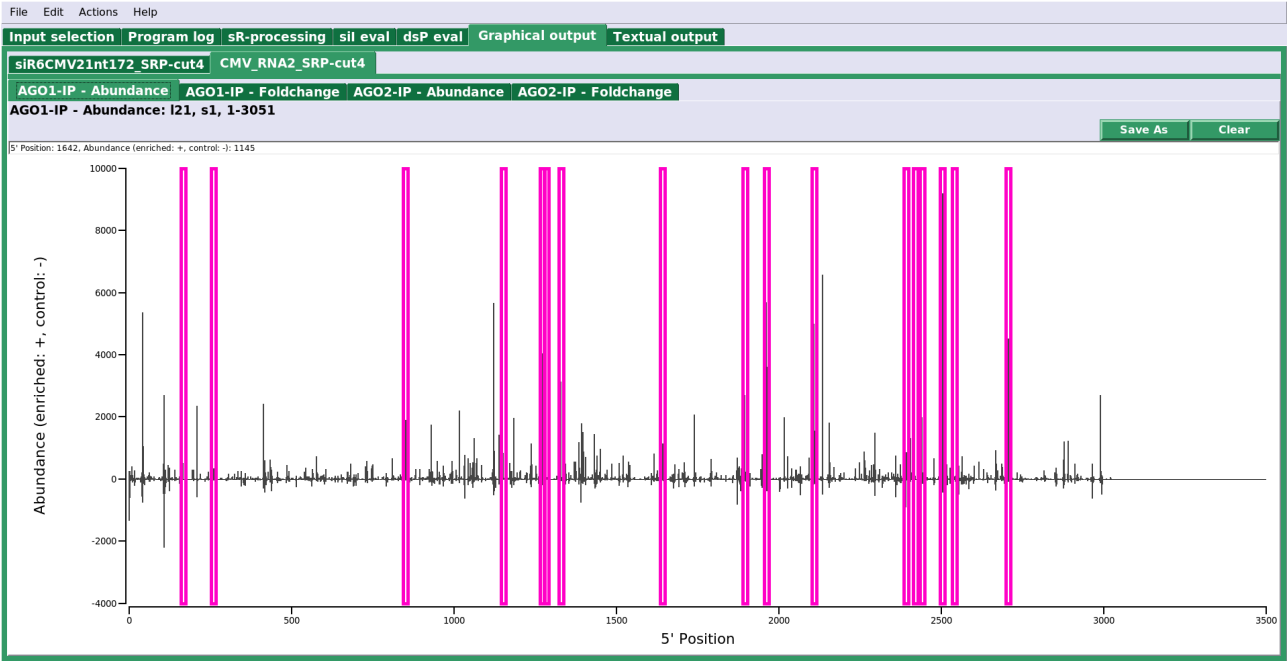


Abundance per 5' position

RNAival plots the abundance of all reads of the selected length as a bar plot, with positive values for the RNA-enriched sample and negative values for the control. Each position corresponds to the 5' position of an RNA of the specified length. As such, the enriched and control values for each position can be compared to identify highly overabundant RNAs. However, this comparison is already covered in a more accessible manner by the volcano-like plot above. Since these two plots are linked, the RNAs selected in the volcano-like plot will also be selected here, allowing manual

observation of the differences in abundance. One can also investigate the distribution of the best candidates over the target RNA, in search of region with a higher density of potent candidate RNAs.

In this example, the best candidates seem to be mostly equally distributed over the available space, with a slight density increase around position 2500.



Detailed information on RNAs

RNAiVal outputs information on the selected candidates into the text output tab.

With the sequences of the candidates readily available, their cleavage efficiency can be tested and dsRNA can be constructed from them.

File Edit Actions Help									
Input selection Program log sR-processing sil eval dsP eval Graphical output Textual output									
siR6CMV21nt172 SRP-cut4 CMV_RNA2_SRP-cut4									
AGO1-IP - Abundance AGO1-IP - Foldchange AGO2-IP - Abundance AGO2-IP - Foldchange									
AGO1-IP - Abundance: I21, s1, 1-3051									
Save As Clear									
5' Position: 1642, Abundance (enriched: +, control: -): 1145									
#Source	ID	3' Pos	Cut Pos	5' Pos	siRNA seq	binding seq	+ -	FC	score
AGO1-IP	siRNA-2524	2504	2514-2515	2524	TTCCCGTCGATTCTGTTTGTG	CACAAACAGAATCGACGGGAA	9203 374	24.544	0.797
AGO1-IP	siRNA-2129	2109	2119-2120	2129	TTGTCAGAATAGGGAATTTTC	GAAAATTCCTATTCTGACAA	4996 219	22.714	0.78
AGO1-IP	siRNA-2727	2707	2717-2718	2727	TTACCGGCGAACCAATCTGTA	TACAGATTGGTTCGCCGGTAA	4525 209	21.552	0.77
AGO1-IP	siRNA-1349	1329	1339-1340	1349	TAAGAAAGACCTGAGGAGCTTC	GAAGTCCTCAGGTCTTTCTTA	3133 106	29.29	0.806
AGO1-IP	siRNA-2461	2441	2451-2452	2461	TTGGAGTTTCAGCTTTGTGCAT	ATGACAAACGTCGAACCTCAA	1992 51	38.327	0.823
AGO1-IP	siRNA-280	260	270-271	280	TTCGAGGGGAATGCGCGTCCT	AGGACGCGCATTCCCTCCGAA	344 4	69.0	0.744
AGO1-IP	siRNA-3010	2990	3000-3001	3010	TACCTGAAACTAGCACGTTG	CAACGTGCTAGTTTCAGGTA	2701 119	22.517	0.763
AGO1-IP	siRNA-1304	1284	1294-1295	1304	TTACTAACGACATTGACAAAG	CTTTGTCAATGTCGTTAGTAA	2993 149	19.96	0.749
AGO1-IP	siRNA-1916	1896	1906-1907	1916	TCGCCTGAGAATAAAAGCTTT	AAAGCTTTTATTCTCAGGCGA	2692 125	21.373	0.756
AGO1-IP	siRNA-1292	1272	1282-1283	1292	TTGACAAAGTTACTGGATGCT	AGCATCCAGTAACTTTGTCAA	4038 251	16.028	0.722
AGO1-IP	siRNA-1142	1122	1132-1133	1142	TTCGTTGTACTACTCTCTTG	CAAGAGAGTAGGTACAACGAA	5672 376	15.048	0.717
AGO1-IP	siRNA-1982	1962	1972-1973	1982	TTGAAAAGAGTTGTGAATTTA	TAAATTCACAACCTTTTCAA	5687 425	13.352	0.698
AGO1-IP	siRNA-871	851	861-862	871	TTGGACATTTTTCATATTGAC	GTCAATATGAAAAATGTCCAA	1907 87	21.682	0.746

dsRNA processing

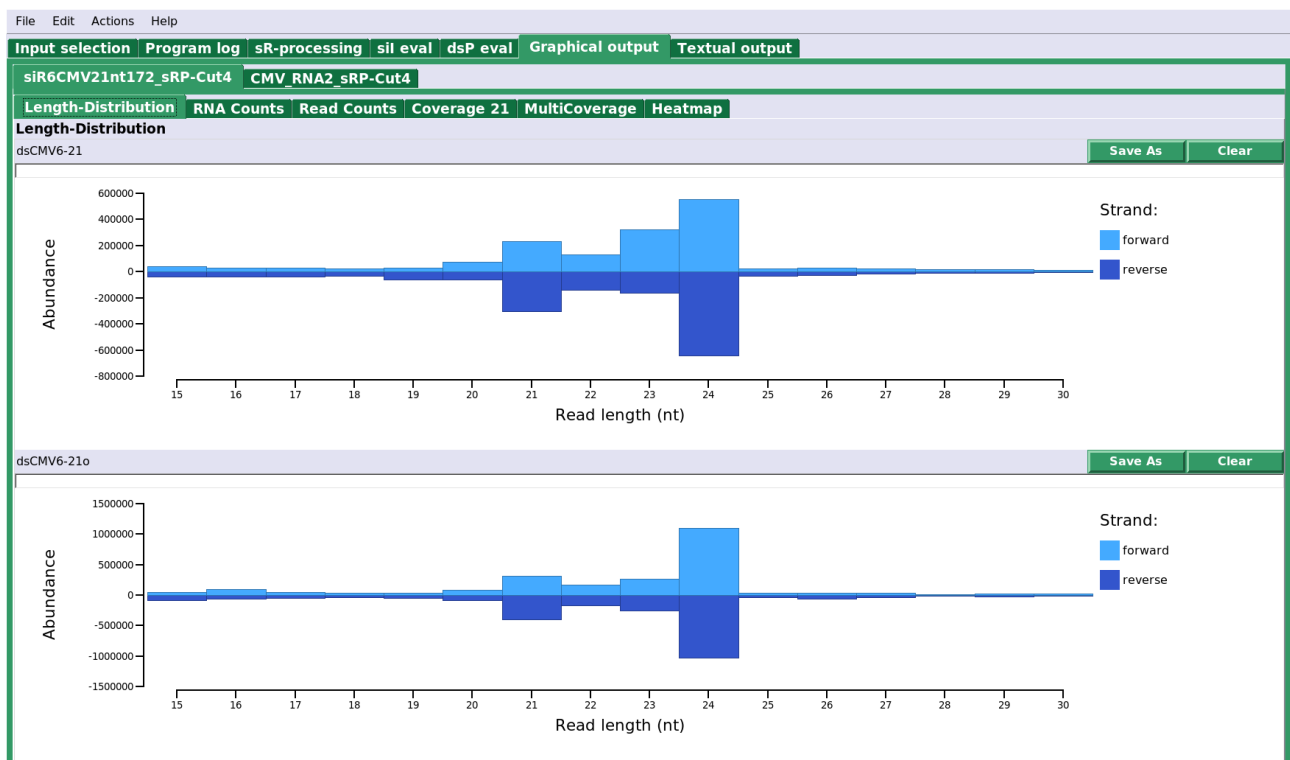
RNAiVal generates various graphs to assess the processing of the input dsRNA.

Read length distribution

Distribution of read lengths observed in the samples. Negative values correspond to reads that map to the reverse strand of the dsRNA.

These plot can be used to compare the amount of reads between different libraries as well as differences in read length abundance.

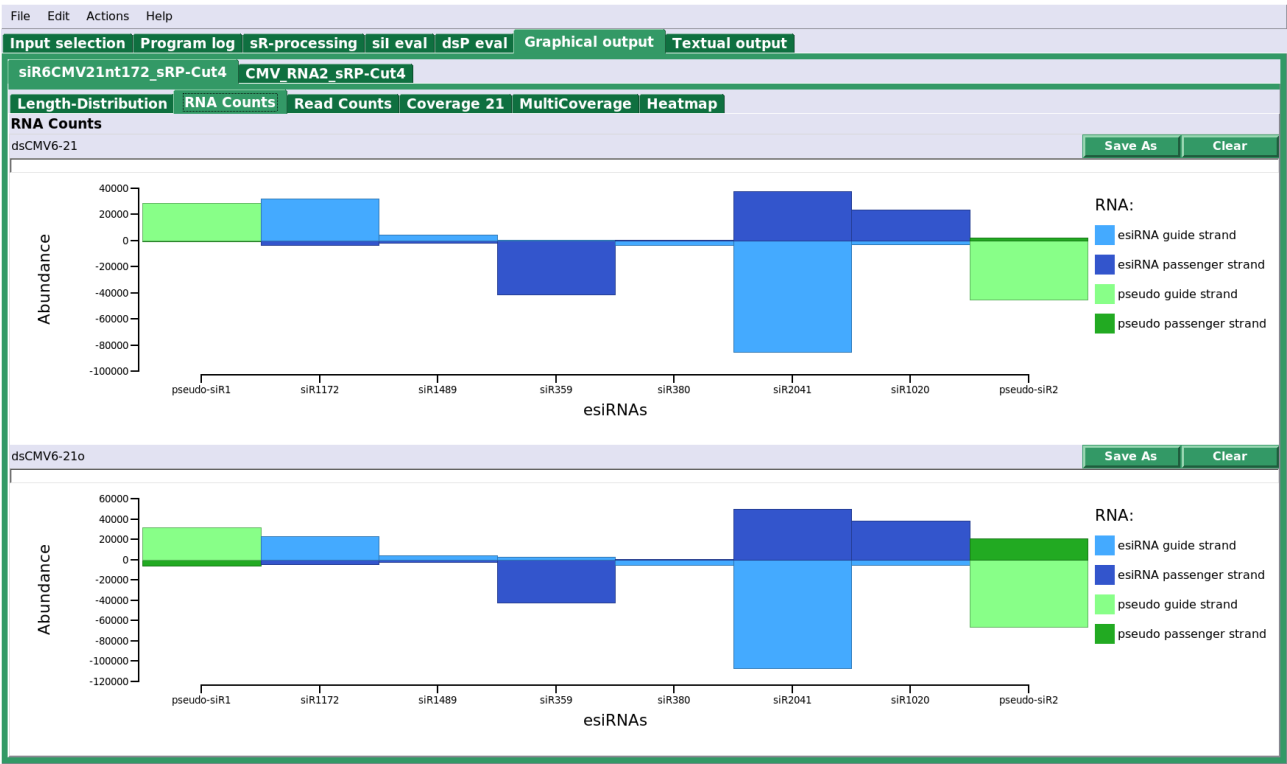
Since these samples are from plants, we expect processing of dsRNA by at least 2 different Dicer-like proteins (DCL): DCL3 cuts dsRNA into 24 nt long siRNA duplexes, DCL4 cuts dsRNA into 21 nt long siRNA duplexes. These plots show us that both of these DCL are active and generate the expected length of reads. It is also visible that more 24 nt long reads exist, which is in line with a higher activity of DCL3 compared to DCL4. It also looks like dsCMV6-21 (top) has a higher relative abundance of 23 nt long reads compared to dsCMV6-21o (bottom). This may indicate a degradation of 24 nt long reads into 23 nt long reads, and possibly even 22 nt long reads. This may be a result of the difference between the two constructs: dsCMV6-21 has blunt ends, while dsCMV6-21o has 2 nt overhangs on each 3' end of the dsRNA.



Annotation abundance

This graph shows only the abundance of the annotated RNAs, giving a quick overview of which RNAs where found more commonly in the samples, as well as whether their guide strands were found in similar amounts.

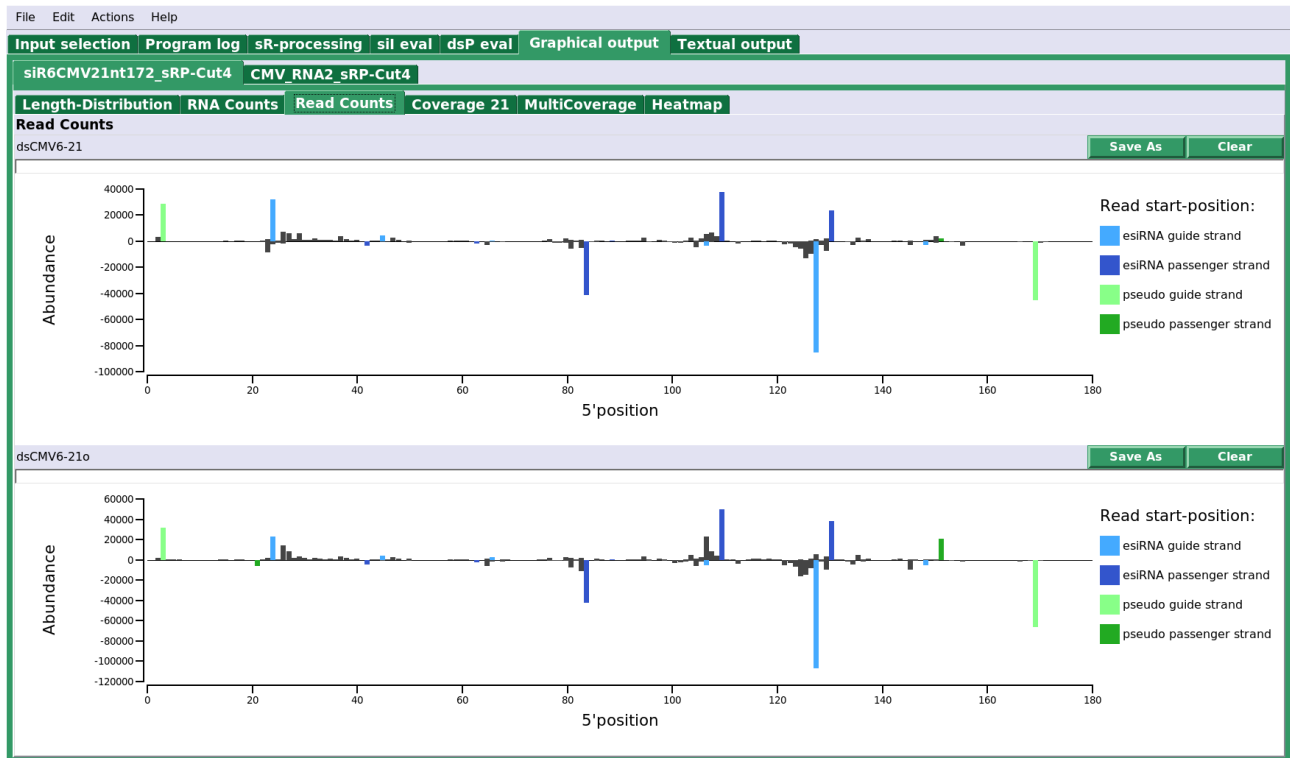
Here we can see a very similar relative expression pattern between the two samples. The main difference is that absence of pseudo guide strands in the dsCMV6-21 (top) sample. However, this is to be expected, since this dsRNA construct has blunt ends and the outer guide strands are thus only 19 nt long, making it impossible for 21 nt long pseudo guide strands to exist.



Abundance per read length

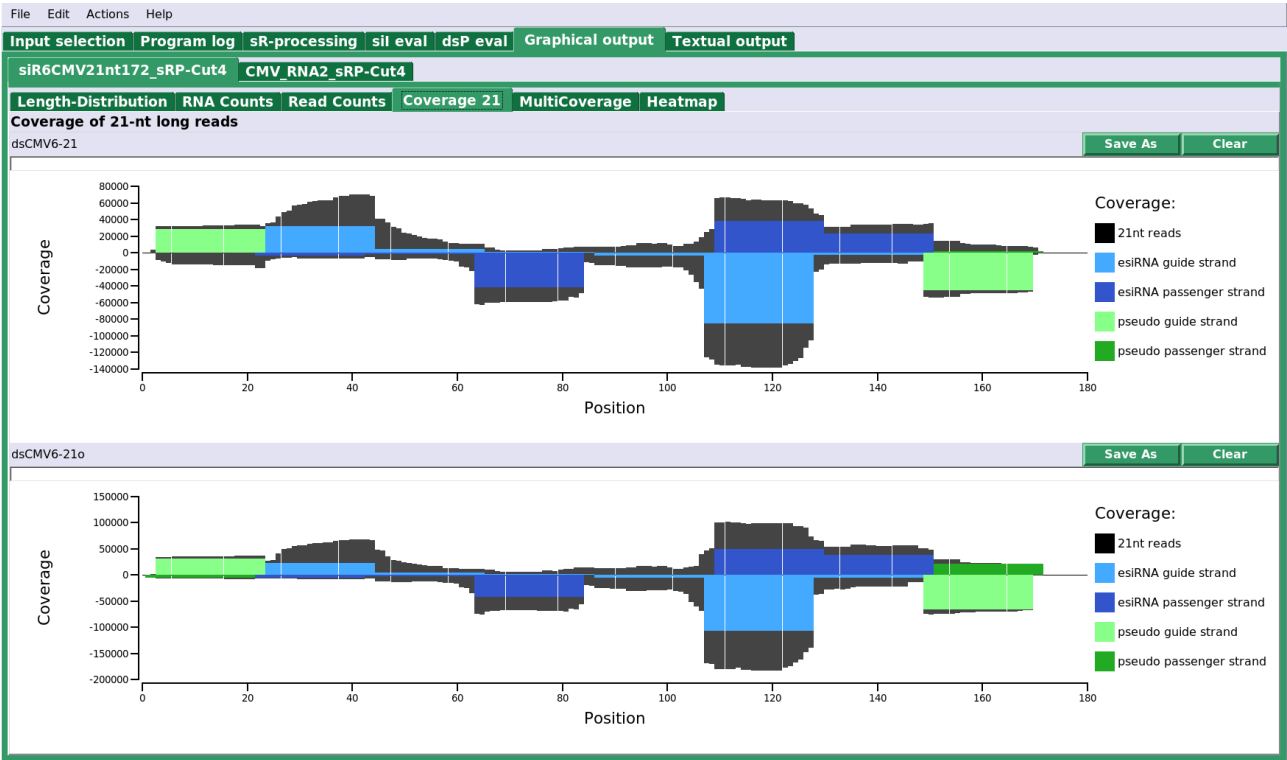
Abundance of each read of a specified length (21 nt in this case); The position corresponds to the 5' position of a read. RNAiVal automatically highlights the the annotated siRNAs in the same colour as in the previous graph.

This shows that most 21 nt long reads correspond to the annotated siRNAs and implies that the dsRNA construct was processed in a mostly regular pattern every 21 nucleotides. Just as the graph above, it highlights the differences in abundance between siRNAs.



Read coverage per length

Coverage of all reads of length 21 over the dsRNA construct. RNAiVal automatically highlights the coverage that comes from reads corresponding to the annotated siRNAs. Same information content as the abundance graph, but better visualises the reads as it also contains information of the structure of the dsRNA cosntruct.

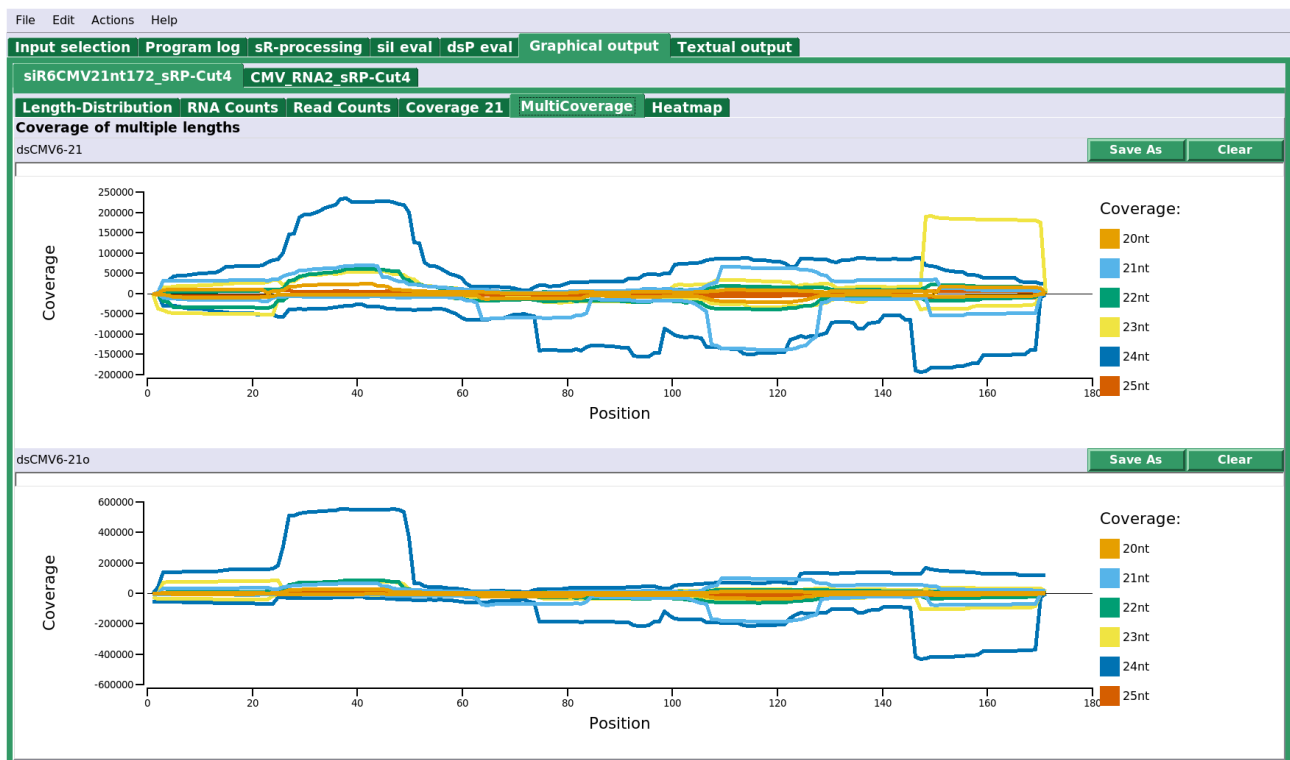


Coverage of multiple lengths

The coverages of reads with different lengths all in the same graph.

Here we can see a higher abundance of 24 nt reads, compared to 21 nt reads. This is in line with the higher activity of DCL3, which cuts 24 nt siRNA duplexes, compared to DCL4, which cuts 21 nt siRNA duplexes.

This graph also shows a strong difference between dsCMV6-21 (top), the blunt-end construct, and dsCMV6-21o (bottom), the 2nt overhang construct: DCL3&4 seem to cut more reliably in a regular interval if the dsRNA has 2nt overhangs. This is in line with previous studies^{2,3}.



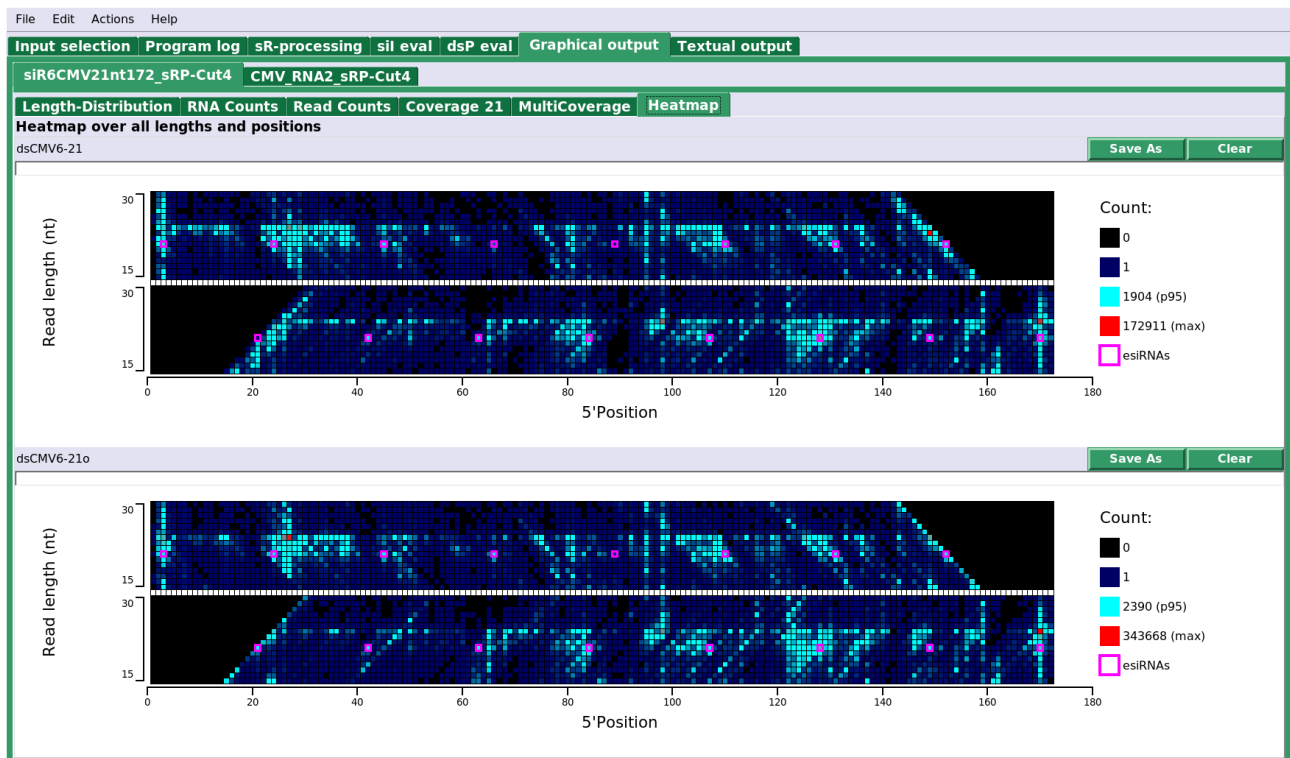
² Nagano H, Fukudome A, Hiraguri A et al. Distinct substrate specificities of Arabidopsis DCL3 and DCL4. *Nucleic Acids Res* 2014;42:1845–56. <https://doi.org/10.1093/nar/gkt1077>

³ Sinha NK, Iwasa J, Shen PS et al. Dicer uses distinct modules for recognizing dsRNA termini. *Science* 2018;359:329–34. <https://doi.org/10.1126/science.aag0921>

Heatmap of all reads

Heatmap over all read lengths and 5' positions on the dsRNA. This heatmap gives an overview over all reads from the samples, making it easier to get a better picture.

For example, vertical and diagonal “lines” or “smears” of read abundance can be seen. These are unexpected and do not correspond to standard processing by DCL. But their existence is intriguing: vertical lines correspond to a set of abundant reads that all have the same 5' position on the target RNA, but different length. This could be due to a degradation of these reads from their 3' ends. Similarly, diagonal lines correspond to sets of abundant reads that all start at incrementing positions, while their length decrement – meaning they all share the same 3' position on the target RNA. This could be due to degradation of these RNAs at their 5' ends.



Filesystem

RNAiVal creates various result files and intermediates and organises them the following way:

Project directory

- Alignments
 - Stores the alignments as BAM files in subdirectories based on the target that was mapped to and the parameterset used (*Alignments/Target/Parameterset/*). Also stores the output of bowtie.
- Counts
 - Stores the counted alignments as tables in subdirectories based on the target that was mapped to and the parameterset used (*Counts/Target/Parameterset/*).
The columns are:
Read-length,
left-most position on the target RNA (5' position for forward match, 3' for reverse matches),
abundance of forward reads from this positions and length,
abundance of reverse reads from this positions and length.
 - Also stores tables of siRNA candidates, sorted by score (descending)
- Fasta
 - Stores fasta versions of embl files used
- Graphics
 - Stores graphical output in subdirectories based on the target that was mapped to and the parameterset used (*Counts/Graphics/Parameterset/*). Files are in SVG format.
- Index
 - Contains all indices created
- Logs
 - Contains logs from all sessions. Can be used to find output from read-processing in a compiled manner.
- Parameters
 - Contains Parameters for all processing and evaluation methods used as json files
- Reads
 - Contains the preprocessing intermediates from sRP in subdirectories based on the Method (sRP) and used parameterset (*Reads/Method/Parameterset/*). They are stored in the directories 0_reads, 1_cutadapt, 2_ngmerge. The Cutadapt and Ngmerge directories also contain the output of the used tools.
 - Once the reads have been preprocessed, intermediates except the ones used for mapping are automatically deleted to save disk space.

- InputFiles.json
 - Stores all information about the read-libraries that were used
- ProjectSettings.json
 - Stores project specific settings

Once reads have been mapped and counted, the evaluation relies only on the count files and the target sequences. As such it is possible to transfer these results without the read files.